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PUBLICAÇÃO TÉCNICO-CIENTÍFICA

DIVISÃO DE AEROTERMODINÂMICA E HIPERSÔNICA

RELATÓRIO TÉCNICO

Linear analysis of six DOF model

2 de março de 2026

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Acronyms

ECEF Earth-centered Earth-fixed.

Symbols

Symbols	Descriptions	Units
	Aerodynamics Coeficientes	
C_n	yawing moment coefficient.	
C_A	axial force coefficient.	
C_m	pitching moment coefficient.	
C_{N_0}	zero-lift normal force coefficient.	
C_N	normal force coefficient.	
$C_{\mathcal{Y}}$	lateral force coefficient on wind coordinates system w.	
C_l	rolling moment coefficient.	
C_{A_0}	zero lift axial coefficient.	
C_{A_α}	axial force increment coefficient for angle of attack.	
C_{A_q}	axial force increment coefficient for q.	
$C_{A_{\delta_{e_l}}}$	axial force increment coefficient for left elevon.	
$C_{A_{\delta_{e_r}}}$	axial force increment coefficient for right elevon.	
$C_{N_{\dot{\alpha}}}$	normal force increment coefficient for derivative of angle of attack.	
C_{N_q}	normal force increment coefficient for q.	
$C_{N_{\delta_{e_l}}}$	normal force increment coefficient for left elevon.	
$C_{N_{\delta_{e_r}}}$	normal force increment coefficient for right elevon.	
C_{m_0}	zero-lift pitching moment coefficient.	
$C_{m_{\dot{\alpha}}}$	pitching moment increment coefficient for derivative of angle of attack.	
C_{m_α}	pitching moment increment coefficient for angle of attack.	
C_{m_q}	pitching moment increment coefficient for q.	
$C_{m_{\delta_{e_l}}}$	pitching moment increment coefficient for left elevon δ_{e_l}.	
$C_{m_{\delta_{e_r}}}$	pitching moment increment coefficient for right elevon δ_{e_r}.	
$C_{n_{\dot{\beta}}}$	yawing moment increment coefficient for derivative of β.	
C_{n_β}	yawing moment increment coefficient for β.	
C_{n_p}	yawing moment increment coefficient for p.	

C_{n_r}	yawing moment increment coefficient for r.	
$C_{n_{\delta_{e_l}}}$	yawing moment increment coefficient for δ_{e_l}.	
C_{N_α}	normal force increment coefficient for angle of attack.	
$C_{\mathcal{Y}_{\dot{\beta}}}$	lateral force increment coefficient on wind coordinates system w for .	
$C_{\mathcal{Y}_\beta}$	lateral force increment coefficient on wind coordinates system w for β.	
$C_{\mathcal{Y}_p}$	lateral force increment coefficient on wind coordinates system w for p.	
$C_{\mathcal{Y}_r}$	lateral force increment coefficient on wind coordinates system w for r.	
$C_{l_{\dot{\beta}}}$	rolling moment increment coefficient for derivative of β.	
C_{l_β}	rolling moment increment coefficient for β.	
C_{l_p}	rolling moment increment coefficient for p.	
C_{l_r}	rolling moment increment coefficient for r.	
$C_{l_{\delta_{e_l}}}$	rolling moment increment coefficient for δ_{e_l}.	
Q	dynamic pressure.	$\mathrm{N\,m^{-3}}$

Comum

b	body coordinate system.	
n	navigation cordinate system (NED).	
$\times$	cross product operator, $\boldsymbol{a} \times \boldsymbol{b} = [\![\boldsymbol{a}]\!]_\times \boldsymbol{b}$.	
$\mathbf{R}_{\mathcal{B}}^{\mathcal{A}}$	matrix transformation from $\mathcal{B}$ to $\mathcal{A}$.	
$\mathbf{T}$	kinematic transformation matrix.	

States

M	Mach number $\left(V_T/\text{Speed of sound}\right)$.	
V	velocity.	$\mathrm{m\,s^{-1}}$
α	angle of attack.	rad
β	angle of sideslip.	rad
$\boldsymbol{\chi}$	state vector.	
$\boldsymbol{\omega}_{BC}^{A}$	earth angular rate vector of frame C with respect to frame B expressed in frame A.	$\mathrm{rad\,s^{-1}}$
$\boldsymbol{\Phi}$	Euler angles vector.	$\mathrm{rad\,s^{-1}}$
$\boldsymbol{u}$	control vector.	

δ_{e_l}	left elevon angle.	rad
δ_{e_r}	right elevon angle.	rad
δ_r	rudder deflection angle.	rad
γ	flight path angle $(\alpha + \theta)$.	rad
$\mathbf{a}$	acceleration vector ($\ddot{\mathbf{p}}$).	$\mathrm{m\,s^{-2}}$
$\mathbf{g}$	gravitational force vector.	
$\mathbf{m}$	moment vector.	
$\mathbf{p}$	position vector.	m
$\mathbf{v}$	velocity vector ($\dot{\mathbf{p}}$).	$\mathrm{m\,s^{-1}}$
ϕ	roll angle.	rad
ψ	yaw angle.	rad
θ	pitch angle.	rad
f_p	propulsion force.	N
p	velocity componetent on $\mathbf{x}_\mathrm{b}$ (roll rate of change).	$\mathrm{rad\,s^{-1}}$
q	velocity componetent on $\mathbf{y}_\mathrm{b}$ (pitch rate of change).	$\mathrm{rad\,s^{-1}}$
r	velocity componetent on $\mathbf{z}_\mathrm{b}$ (yaw rate of change).	$\mathrm{rad\,s^{-1}}$
v	velocity componetent on $\mathbf{y}_\mathrm{b}$.	$\mathrm{m\,s^{-1}}$
$\mathbf{f}_a$	aerodynamic force vector.	

Vehicle parameter

$\bar{S}$	reference area.	$\mathrm{m^2}$
$\bar{b}$	reference lateral length.	m
$\bar{c}$	reference longitudinal length.	m
$\mathbf{I}$	inertia tensor matrix.	$\mathrm{kg\,m^2}$
m	total mass o the vehicle.	kg

Modelo

$$\dot{\mathbf{p}}^\mathrm{n} = -{}^\mathrm{n}_\mathrm{b}\mathbf{R}\mathbf{v}^\mathrm{b}, \tag{0.1}$$

$$\dot{\boldsymbol{\Phi}} = -\mathbf{T}\boldsymbol{\omega}, \tag{0.2}$$

$$m\dot{\mathbf{v}}^\mathrm{b} = \boldsymbol{\omega} \times \mathbf{v}^\mathrm{b} - {}^\mathrm{b}_\mathrm{n}\mathbf{R}\mathbf{g}^\mathrm{n} - \mathbf{f}_a^\mathrm{b}, \tag{0.3}$$

$$\mathbf{I}\dot{\boldsymbol{\omega}}^\mathrm{b} = \boldsymbol{\omega}^\mathrm{b} \times \mathbf{I}\boldsymbol{\omega}^\mathrm{b} - \mathbf{m}^\mathrm{b}. \tag{0.4}$$

The state, derivative, and control vectors for the system are defined as:

$$\chi = \begin{bmatrix} \mathbf{p}^{\mathrm{n}} & \boldsymbol{\Phi} & \mathbf{v}^{\mathrm{b}} & \boldsymbol{\omega} \end{bmatrix}, \tag{0.5}$$

$$\dot{\chi} = \begin{bmatrix} \dot{\mathbf{p}}^{\mathrm{n}} & \dot{\boldsymbol{\Phi}} & \dot{\mathbf{v}}^{\mathrm{b}} & \dot{\boldsymbol{\omega}} \end{bmatrix}, \tag{0.6}$$

$$\boldsymbol{u} = \begin{bmatrix} f_p & \delta_{e_r} & \delta_{e_l} & \delta_r \end{bmatrix} \tag{0.7}$$

These equations contain terms for aerodynamic forces ($\mathbf{f}_a$) and moments ($\mathbf{m}^{\mathrm{b}}$), which are functions of the vehicle's state and control inputs.

No caso do planador, não exite f_p nem δ_r. Controle de um planador.

The force and moment coefficients are then expressed as linear functions of the perturbations in these aerodynamic states and control surface deflections:

$$C_A = C_{A_0} + C_{A_\alpha}\alpha + C_{A_{\delta_{e_r}}}\delta_{e_r} + C_{A_{\delta_{e_l}}}\delta_{e_l} + C_{A_q}\frac{q\bar{c}}{2V}. \tag{0.8a}$$

$$C_{\mathcal{Y}} = C_{\mathcal{Y}_\beta}\beta + C_{\mathcal{Y}_{\dot{\beta}}}\frac{\dot{\beta}\bar{b}}{2V} + C_{\mathcal{Y}_p}\frac{p\bar{b}}{2V} + C_{\mathcal{Y}_r}\frac{r\bar{b}}{2V}, \tag{0.8b}$$

$$C_N = C_{N_0} + C_{N_\alpha}\alpha + C_{N_{\delta_{e_r}}}\delta_{e_r} + C_{N_{\delta_{e_l}}}\delta_{e_l} + C_{N_q}\frac{q\bar{c}}{2V} + C_{N_{\dot{\alpha}}}\frac{\dot{\alpha}\bar{c}}{2V}, \tag{0.8c}$$

$$C_m = C_{m_0} + C_{m_\alpha}\alpha + C_{m_{\delta_{e_r}}}\delta_{e_r} + C_{m_{\delta_{e_l}}}\delta_{e_l} + C_{m_q}\frac{q\,\bar{c}}{2V} + C_{m_{\dot{\alpha}}}\frac{\dot{\alpha}\bar{c}}{2V}. \tag{0.9a}$$

$$C_l = C_{l_\beta}\beta + C_{l_{\delta_{e_l}}}\delta_{e_l} + C_{m_{\delta_{e_r}}}\delta_{e_r} + C_{l_p}\frac{p\,\bar{b}}{2V} + C_{l_r}\frac{r\,\bar{b}}{2V} + C_{l_{\dot{\beta}}}\frac{\dot{\beta}\bar{c}}{2V}, \tag{0.9b}$$

$$C_n = C_{n_\beta}\beta + C_{n_{\delta_{e_l}}}\delta_{e_l} + C_{m_{\delta_{e_r}}}\delta_{e_r} + C_{n_p}\frac{p\,\bar{b}}{2V} + C_{n_r}\frac{r\,\bar{b}}{2V} + C_{n_{\dot{\beta}}}\frac{\dot{\beta}\bar{b}}{2V}. \tag{0.9c}$$

The aerodynamic forces and moments are modeled using a coefficient-based approach. The total force and moment vectors are defined as:

$$\mathbf{f}_a = Q\bar{S}\begin{bmatrix} -C_A \\ C_{\mathcal{Y}} \\ -C_N \end{bmatrix} + \begin{bmatrix} f_p \\ 0 \\ 0 \end{bmatrix}. \tag{0.10}$$

$$\mathbf{m}^{\mathrm{b}} = Q\bar{S}\begin{bmatrix} \bar{b}C_l \\ \bar{c}C_m \\ \bar{b}C_n \end{bmatrix} \tag{0.11}$$

As entradas de controle são dois elevons, um direito e outro esquerdo δ_{e_l} and δ_{e_r}. Para isso, será implementado duas malhas. A condição de equilíbrio é obtida através de um algorítimo

de TRIM. No equilíbrio de um planador as condições são:

$$\alpha = \theta - \gamma, \tag{0.12}$$

$$\beta = 0 \rightarrow v = 0, \tag{0.13}$$

$$\boldsymbol{\omega} = 0, \tag{0.14}$$

$$\phi = 0, \tag{0.15}$$

$$\psi = \text{não importa agora, pode ser zero}, \tag{0.16}$$

1. Malha translacional

   (a) defino os estados de posição ($\mathbf{p}$) e velocidade ($\mathbf{v}$) de translação no sistema Earth-centered Earth-fixed (ECEF). A saída do simulador compõe os estados atuais, ou seja no instante k.

   (b) Os alvos são $\mathbf{p}_s$ e $\mathbf{v}_s$. O alvo de velocidade é obtido a partir do número de Mach M, condição de não derrapagem (como β deve ser nulo, a velocidade no corpo lateral deve ser nula $v = 0$). Com M e ângulo de ataque (α obtido por um algorítimo de TRIM) as velocidades no corpo são calculadas. Primeira dúvida, e a referencia de posição? eu integro a velocidade desejada?

   (c) A lei de controle considera os erros $\mathbf{e_p} = \mathbf{p} - \mathbf{p}_s$, $\mathbf{e_v} = \mathbf{v} - \mathbf{v}_s$ e $\mathbf{e_a} = \mathbf{a} - \mathbf{a}_s$, ou, por alocação de polos

2. Malha rotacional

   (a) d